CLASSROOM TIME TRIALS & COMPETITION

StackCalc Calculator Applications

UIL-Style speed and accuracy proficiency tests aligned to maximum stack depth limits.

1. INVITATIONAL TIME TRIAL & SPEED SCORING RULES

Modeled after University Interscholastic League (UIL) Calculator Applications competitions, these timed trials challenge students to build rapid, instinctive fluency with Reverse Polish Notation (RPN). Each sprint tests numerical accuracy, operand sequencing, and strict stack depth consciousness.

CORRECT ANSWER	INCORRECT / ERROR	UNATTEMPTED / SKIP	SPEED BONUS
+5 Points	**-2 Points**	**0 Points**	**+1 pt / rem. min**

2. MAXIMUM STACK LEVEL DISCIPLINE

Competitors must respect the Max Stack Depth badge on each trial. Problems must be solved using only the allowed stack registers (Stack Level 1 = single register X; Stack Level 2 = registers X & Y; Stack Level 4 = full 4-register stack T-Z-Y-X). Solutions requiring deeper stacks are disqualified.

3. OFFICIAL COMPETITOR TIME TRIAL LOG SHEET

Trial	Topic / Stage	Max Stack	Time	Attempt	Correct	Errors	
T-1	Unit 1 Time Trial: Step Counter & Accumu	**Level 1**	5 min	___/10	___(+5)	___(-2)	_____
T-2	Unit 2 Time Trial: Part-Part-Whole & Sta	**Level 2**	5 min	___/10	___(+5)	___(-2)	_____
T-3	Unit 3 Time Trial: Ten-Frames & Compleme	**Level 2**	5 min	___/10	___(+5)	___(-2)	_____
T-4	Unit 4 Time Trial: Number Racks, Doubles	**Level 2**	5 min	___/10	___(+5)	___(-2)	_____
T-CUM_1	Cumulative Trial C1: Primary Stack Funda	**Level 2**	10 min	___/10	___(+5)	___(-2)	_____
T-5	Unit 5 Time Trial: 4-Register Stack & Fo	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-6	Unit 6 Time Trial: Fraction Studio & Mix	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-7	Unit 7 Time Trial: Expression Trees & Po	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-8	Unit 8 Time Trial: Coordinate Lab & Pyth	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-CUM_2	Cumulative Trial C2: Intermediate Stack	**Level 4**	15 min	___/10	___(+5)	___(-2)	_____
T-9	Unit 9 Time Trial: Trigonometry, Vectors	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-10	Unit 10 Time Trial: Exponential Growth &	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-11	Unit 11 Time Trial: Combinatorics, Stats	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-12	Unit 12 Time Trial: Digital Logic, Base	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-CUM_3	Cumulative Trial C3: Grand RPN Invitatio	**Level 4**	20 min	___/10	___(+5)	___(-2)	_____

Classroom Implementation

Run time trials as weekly sprint rounds. Time starts when students turn the page. All keystrokes must be entered on the physical StackCalc hardware or firmware simulator.

STUDENT CALCULATOR APPLICATIONS

Unit 1 Time Trial: Step Counter & Accumulator Spr

Foundations of Counting, Cardinality & The One-More Step Counter • Stage: Ten-Frame Stage (4x2)

MAX STACK DEPTH: 1 | **TIME LIMIT: 5 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**1 + 1 + 1 (Count 3 steps)** Stack Hint: Single register accumulator	Keystroke Entry:	**Final Answer:**
2	**5 + 1 + 1 (Start at 5, step 2)** Stack Hint: Accumulate in X	Keystroke Entry:	**Final Answer:**
3	**Step count 4 items: start 0, step +1** Stack Hint: Repeat + key	Keystroke Entry:	**Final Answer:**
4	**Count 8 items with auto-lift** Stack Hint: Continuous + entry	Keystroke Entry:	**Final Answer:**
5	**Count down: start 6, step -1, step -1** Stack Hint: Subtract step	Keystroke Entry:	**Final Answer:**
6	**Clear and count: C then 1+1+1+1+1** Stack Hint: Clear before count	Keystroke Entry:	**Final Answer:**
7	**Count by 2s: 2 + 2 + 2** Stack Hint: Skip count by 2	Keystroke Entry:	**Final Answer:**
8	**Step count to ten: 7 + 1 + 1 + 1** Stack Hint: Reach upper frame	Keystroke Entry:	**Final Answer:**
9	**Zero set: 0 + 9** Stack Hint: Identity of zero	Keystroke Entry:	**Final Answer:**
10	**Count back from 10: 10 - 1 - 1 - 1** Stack Hint: Decrement from 10	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 1 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

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STUDENT CALCULATOR APPLICATIONS

Unit 2 Time Trial: Part-Part-Whole & Stack Additio

Number Bonds to 10 & 20 via Stack Registers Y and X • Stage: Number Bond Stage (3x2)

MAX STACK DEPTH: 2 | **TIME LIMIT: 5 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**Part 4, Part 3 -> Whole?** Stack Hint: Y=4, X=3, [+] combines	Keystroke Entry:	**Final Answer:**
2	**Part 5, Part 5 -> Whole?** Stack Hint: Ten bond	Keystroke Entry:	**Final Answer:**
3	**Whole 9, Part 6 -> Missing Part?** Stack Hint: Decompose whole 9	Keystroke Entry:	**Final Answer:**
4	**Whole 10, Part 4 -> Missing Part?** Stack Hint: Decompose whole 10	Keystroke Entry:	**Final Answer:**
5	**Part 8, Part 7 -> Whole?** Stack Hint: Sum past ten	Keystroke Entry:	**Final Answer:**
6	**Whole 14, Part 8 -> Missing Part?** Stack Hint: Decompose 14	Keystroke Entry:	**Final Answer:**
7	**Part 6, Part 7 -> Whole?** Stack Hint: Part combination	Keystroke Entry:	**Final Answer:**
8	**Whole 18, Part 9 -> Missing Part?** Stack Hint: Double decomposition	Keystroke Entry:	**Final Answer:**
9	**Part 12, Part 8 -> Whole?** Stack Hint: Full twenty bond	Keystroke Entry:	**Final Answer:**
10	**Whole 25, Part 11 -> Missing Part?** Stack Hint: Two-digit decompose	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

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STUDENT CALCULATOR APPLICATIONS

Unit 3 Time Trial: Ten-Frames & Complements to 1

5-Structure, Complements & Bridging Through 10 • Stage: Ten-Frame Stage (4x2)

MAX STACK DEPTH: 2 | TIME LIMIT: 5 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**Complement of 7 to 10** Stack Hint: 10 in Y, 7 in X	Keystroke Entry:	**Final Answer:**
2	**Complement of 4 to 10** Stack Hint: 10 in Y, 4 in X	Keystroke Entry:	**Final Answer:**
3	**Bridge: 8 + 5 = 8 + 2 + 3** Stack Hint: Bridge through 10	Keystroke Entry:	**Final Answer:**
4	**Bridge: 9 + 6 = 9 + 1 + 5** Stack Hint: Bridge through 10	Keystroke Entry:	**Final Answer:**
5	**Bridge: 7 + 6 = 7 + 3 + 3** Stack Hint: Bridge through 10	Keystroke Entry:	**Final Answer:**
6	**Complement of 1 to 10** Stack Hint: 10 in Y, 1 in X	Keystroke Entry:	**Final Answer:**
7	**Double frame: 20 - 7** Stack Hint: Subtract across 2 frames	Keystroke Entry:	**Final Answer:**
8	**Bridge: 8 + 7 = 8 + 2 + 5** Stack Hint: Bridge through 10	Keystroke Entry:	**Final Answer:**
9	**Place value: 10 + 6** Stack Hint: Teen number formation	Keystroke Entry:	**Final Answer:**
10	**Frame difference: 10 - 2** Stack Hint: Ten complement	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

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STUDENT CALCULATOR APPLICATIONS

Unit 4 Time Trial: Number Racks, Doubles & Group

5-Structure, Automatic Lift Duplication & Early Multiplication • Stage: Number Rack (4x2)

MAX STACK DEPTH: 2 | **TIME LIMIT: 5 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**Double 6: 6 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Entry:	**Final Answer:**
2	**Double 8: 8 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Entry:	**Final Answer:**
3	**Near-Double: 7 + 8** Stack Hint: Double 7 plus 1	Keystroke Entry:	**Final Answer:**
4	**Near-Double: 6 + 7** Stack Hint: Double 6 plus 1	Keystroke Entry:	**Final Answer:**
5	**Double 9: 9 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Entry:	**Final Answer:**
6	**Near-Double: 8 + 9** Stack Hint: Double 8 plus 1	Keystroke Entry:	**Final Answer:**
7	**Equal groups: 3 groups of 4** Stack Hint: Multiply group size	Keystroke Entry:	**Final Answer:**
8	**Equal groups: 4 groups of 5** Stack Hint: Multiply group size	Keystroke Entry:	**Final Answer:**
9	**Double 12: 12 ENTER +** Stack Hint: ENTER duplicates to Y	Keystroke Entry:	**Final Answer:**
10	**Near-Double: 9 + 10** Stack Hint: Double 9 plus 1	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

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STUDENT CALCULATOR APPLICATIONS

Cumulative Trial C1: Primary Stack Fundamentals

Comprehensive Time Trial across Units 1–4 (Max Stack Depth 2) • Stage: Gridfinity 4x4 Primary Desk Dock

MAX STACK DEPTH: 2 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**Step count 7 items from 1** Stack Hint: Accumulate in X	Keystroke Entry:	**Final Answer:**
2	**Number bond: 9 and 7 -> Whole?** Stack Hint: Y=9, X=7	Keystroke Entry:	**Final Answer:**
3	**Decompose: Whole 17, Part 8** Stack Hint: 17 in Y, 8 in X	Keystroke Entry:	**Final Answer:**
4	**Complement of 3 to 10** Stack Hint: Ten complement	Keystroke Entry:	**Final Answer:**
5	**Double 14: 14 ENTER +** Stack Hint: Duplicate to Y	Keystroke Entry:	**Final Answer:**
6	**Near-double: 11 + 12** Stack Hint: Double 11 plus 1	Keystroke Entry:	**Final Answer:**
7	**Bridge through 10: 9 + 8** Stack Hint: Bridge with frame	Keystroke Entry:	**Final Answer:**
8	**Equal groups: 6 groups of 5** Stack Hint: 5 in Y, 6 in X	Keystroke Entry:	**Final Answer:**
9	**Decompose: Whole 30, Part 12** Stack Hint: Missing part	Keystroke Entry:	**Final Answer:**
10	**Combined: Double 8 then add 5** Stack Hint: Chained operations	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 2 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

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STUDENT CALCULATOR APPLICATIONS

Unit 5 Time Trial: 4-Register Stack & Four Operati

T-Z-Y-X Flow, Automatic Lift & Parenthesis-Free Arithmetic • Stage: Stack Stage (3x3)

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**(18 - 6) × 3** Stack Hint: Auto-lift pushes 12 to Y	Keystroke Entry:	**Final Answer:**
2	**(24 + 16) ÷ 8** Stack Hint: Sum in Y, divide by 8	Keystroke Entry:	**Final Answer:**
3	**45 ÷ (12 - 3)** Stack Hint: Push 45 to Y, sub in X	Keystroke Entry:	**Final Answer:**
4	**(15 × 4) + (18 ÷ 2)** Stack Hint: Two branch accumulation	Keystroke Entry:	**Final Answer:**
5	**(50 - 14) ÷ (3 + 6)** Stack Hint: Stack holds numerator in Y	Keystroke Entry:	**Final Answer:**
6	**12 × 5 - 8 × 4** Stack Hint: Two branch subtraction	Keystroke Entry:	**Final Answer:**
7	**((20 - 8) × 4) ÷ 6** Stack Hint: Chained pipeline	Keystroke Entry:	**Final Answer:**
8	**72 ÷ (4 + 5) × 3** Stack Hint: Parenthesis-free stack	Keystroke Entry:	**Final Answer:**
9	**100 - (16 × 5 - 12)** Stack Hint: Deep stack evaluate	Keystroke Entry:	**Final Answer:**
10	**(8 + 4) × (9 - 3) ÷ 12** Stack Hint: Multi-step evaluate	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

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STUDENT CALCULATOR APPLICATIONS

Unit 6 Time Trial: Fraction Studio & Mixed Number

Two-Dot Notation, Reciprocals & Height Equivalence • Stage: Fraction Studio (4x2)

MAX STACK DEPTH: 4 | TIME LIMIT: 10 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**1/2 + 1/4 (Two-dot entry)** Stack Hint: Decimal equivalent 0.75	Keystroke Entry:	**Final Answer:**
2	**3/4 - 1/2** Stack Hint: Decimal equivalent 0.25	Keystroke Entry:	**Final Answer:**
3	**Mixed: 2 1/2 + 1 1/4** Stack Hint: Mixed number addition	Keystroke Entry:	**Final Answer:**
4	**Reciprocal of 4/5: .4.5 [1/x]** Stack Hint: Reciprocal key	Keystroke Entry:	**Final Answer:**
5	**3/8 + 1/4** Stack Hint: Three eighths plus fourth	Keystroke Entry:	**Final Answer:**
6	**2/3 × 3/4** Stack Hint: Fraction multiplication	Keystroke Entry:	**Final Answer:**
7	**3/4 ÷ 1/2** Stack Hint: Fraction division	Keystroke Entry:	**Final Answer:**
8	**1 1/2 × 4** Stack Hint: Mixed scaling	Keystroke Entry:	**Final Answer:**
9	**5 - 2 3/4** Stack Hint: Subtract mixed number	Keystroke Entry:	**Final Answer:**
10	**(1/3 + 1/6) × 12** Stack Hint: Combined fraction pipeline	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STACKCALC / LEARNING LAB

STUDENT CALCULATOR APPLICATIONS

Unit 7 Time Trial: Expression Trees & Postorder Tr

Binary Syntax Trees, Postorder Traversal & Parenthesis Elimination • Stage: Expression Tree (4x3)

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**(8 + 2) × (9 - 4)** Stack Hint: Left-Right-Multiply	Keystroke Entry:	**Final Answer:**
2	**((14 - 6) × 3) + (24 ÷ 4)** Stack Hint: Two subtree combine	Keystroke Entry:	**Final Answer:**
3	**(30 - (4 × 5)) × (12 ÷ (2 + 4))** Stack Hint: Deep nested subtrees	Keystroke Entry:	**Final Answer:**
4	**(5 + 3) × (8 - 2) ÷ (6 - 2)** Stack Hint: Three branch traversal	Keystroke Entry:	**Final Answer:**
5	**((18 ÷ 3) + 4) × ((15 - 7) ÷ 2)** Stack Hint: Postorder multiplication	Keystroke Entry:	**Final Answer:**
6	**(10 + 5 × 2) ÷ (12 - 2 × 4)** Stack Hint: Eliminating PEMDAS ambiguity	Keystroke Entry:	**Final Answer:**
7	**(25 - 5 × 3) × (4 + 6 ÷ 2)** Stack Hint: Nested tree traversal	Keystroke Entry:	**Final Answer:**
8	**((7 + 3) × 4 - 8) ÷ 4** Stack Hint: Chained operations	Keystroke Entry:	**Final Answer:**
9	**(48 ÷ (2 + 6)) × (15 - 3 × 3)** Stack Hint: Tree sub-expression	Keystroke Entry:	**Final Answer:**
10	**(16 - (4 + 2)) × (18 ÷ (3 × 2))** Stack Hint: Stack tree depth 4	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

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STUDENT CALCULATOR APPLICATIONS

Unit 8 Time Trial: Coordinate Lab & Pythagorean D

Cartesian Pegboards, Euclidean Distance Pipeline & Slope • Stage: Coordinate Lab (4x4)

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**□x = 3, □y = 4 -> Distance d** Stack Hint: Pythagorean triple (3,4,5)	Keystroke Entry:	**Final Answer:**
2	**□x = 6, □y = 8 -> Distance d** Stack Hint: Scaled triple (6,8,10)	Keystroke Entry:	**Final Answer:**
3	**□x = 5, □y = 12 -> Distance d** Stack Hint: Pythagorean triple (5,12,13)	Keystroke Entry:	**Final Answer:**
4	**P1(1, 2) to P2(4, 6) -> Distance** Stack Hint: Full distance formula	Keystroke Entry:	**Final Answer:**
5	**P1(2, 3) to P2(8, 11) -> Distance** Stack Hint: Full distance formula	Keystroke Entry:	**Final Answer:**
6	**Slope m: Rise 12, Run 4** Stack Hint: Rise in Y, Run in X	Keystroke Entry:	**Final Answer:**
7	**Slope m: P1(2, 5) to P2(6, 17)** Stack Hint: □y ÷ □x	Keystroke Entry:	**Final Answer:**
8	**Hypotenuse: legs 9 and 12** Stack Hint: Leg accumulation	Keystroke Entry:	**Final Answer:**
9	**□x = 8, □y = 15 -> Distance d** Stack Hint: Triple (8,15,17)	Keystroke Entry:	**Final Answer:**
10	**P1(0, 0) to P2(7, 24) -> Distance** Stack Hint: Triple (7,24,25)	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Cumulative Trial C2: Intermediate Stack Mastery C

Comprehensive Time Trial across Units 1–8 (Max Stack Depth 4) • Stage: Stack Stage & Expression Tree Hub

MAX STACK DEPTH: 4 | **TIME LIMIT: 15 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**(32 - 14) × 5 ÷ 6** Stack Hint: Auto-lift chaining	Keystroke Entry:	**Final Answer:**
2	**3/4 + 1/2 × 1/4 (order of ops)** Stack Hint: Fractions on stack	Keystroke Entry:	**Final Answer:**
3	**Distance: P1(3, 4) to P2(9, 12)** Stack Hint: Pythagorean distance	Keystroke Entry:	**Final Answer:**
4	**Mixed: 3 1/2 - 1 3/4** Stack Hint: Two-dot fraction subtraction	Keystroke Entry:	**Final Answer:**
5	**Expression tree: ((15+5)×(8-3))÷10** Stack Hint: Tree postorder	Keystroke Entry:	**Final Answer:**
6	**Double 18 then sub 9** Stack Hint: Duplicate to Y	Keystroke Entry:	**Final Answer:**
7	**Slope: P1(-1, -2) to P2(3, 10)** Stack Hint: Negative coordinates	Keystroke Entry:	**Final Answer:**
8	**Four operations: 80 - 15 × 4 + 12** Stack Hint: RPN stack evaluation	Keystroke Entry:	**Final Answer:**
9	**Reciprocal sum: 1/(1/4 + 1/6)** Stack Hint: Reciprocal pipeline	Keystroke Entry:	**Final Answer:**
10	**Distance from origin to (12, 16)** Stack Hint: Radial distance	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STACKCALC / LEARNING LAB

STUDENT CALCULATOR APPLICATIONS

Unit 9 Time Trial: Trigonometry, Vectors & Polar P

Right Triangles, Polar Conversions & Vector Decomposition • Stage: Coordinate Lab (4x4)

MAX STACK DEPTH: 4 | TIME LIMIT: 10 MIN | MAX SCORE: 50 PTS

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**sin(30°)** Stack Hint: Basic sine ratio	Keystroke Entry:	**Final Answer:**
2	**cos(60°)** Stack Hint: Basic cosine ratio	Keystroke Entry:	**Final Answer:**
3	**tan(45°)** Stack Hint: Tangent ratio	Keystroke Entry:	**Final Answer:**
4	**Hypotenuse 10, angle 30° -> Opposite (y)** Stack Hint: Opposite = h × sin(□)	Keystroke Entry:	**Final Answer:**
5	**Adjacent (x): Hypotenuse 12, angle 60°** Stack Hint: Adjacent = h × cos(□)	Keystroke Entry:	**Final Answer:**
6	**Polar to Rect: r=10, □=0° -> x** Stack Hint: Polar conversion key	Keystroke Entry:	**Final Answer:**
7	**Rect to Polar: x=3, y=4 -> magnitude r** Stack Hint: Vector magnitude	Keystroke Entry:	**Final Answer:**
8	**Rect to Polar: x=5, y=12 -> magnitude r** Stack Hint: Vector magnitude	Keystroke Entry:	**Final Answer:**
9	**Vector magnitude: components 8 and 6** Stack Hint: Polar magnitude	Keystroke Entry:	**Final Answer:**
10	**asin(0.5) in degrees** Stack Hint: Inverse sine	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STACKCALC / LEARNING LAB

STUDENT CALCULATOR APPLICATIONS

Unit 10 Time Trial: Exponential Growth & Logarith

Natural Base e, Continuous Compounding & Doubling Time • Stage: Stack Stage (3x3)

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**2^5 (Power key)** Stack Hint: y to power x	Keystroke Entry:	**Final Answer:**
2	**3^4 (Power key)** Stack Hint: y to power x	Keystroke Entry:	**Final Answer:**
3	**ln(e)** Stack Hint: Natural log inverse	Keystroke Entry:	**Final Answer:**
4	**e^0** Stack Hint: Base e exponent	Keystroke Entry:	**Final Answer:**
5	**log10(1000)** Stack Hint: Common log	Keystroke Entry:	**Final Answer:**
6	**Compound: 1000 × (1.05)^2** Stack Hint: Annual compounding	Keystroke Entry:	**Final Answer:**
7	**Doubling time: ln(2) ÷ 0.07** Stack Hint: Continuous doubling rule	Keystroke Entry:	**Final Answer:**
8	**Half-life: 100 × (0.5)^3** Stack Hint: Decay model	Keystroke Entry:	**Final Answer:**
9	**Continuous: 500 × e^(0.04 × 5)** Stack Hint: P e^(rt) formula	Keystroke Entry:	**Final Answer:**
10	**2^10** Stack Hint: Powers of two	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STACKCALC / LEARNING LAB

STUDENT CALCULATOR APPLICATIONS

Unit 11 Time Trial: Combinatorics, Stats & Probabi

Factorials, Permutations, Combinations & [Σ+] Accumulation • Stage: Coordinate Lab (4x4)

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**5! (Factorial)** Stack Hint: x factorial key	Keystroke Entry:	**Final Answer:**
2	**6! (Factorial)** Stack Hint: x factorial key	Keystroke Entry:	**Final Answer:**
3	**Permutations P(5, 2) = 5! / 3!** Stack Hint: Stack permutation	Keystroke Entry:	**Final Answer:**
4	**Permutations P(6, 3) = 6! / 3!** Stack Hint: Stack permutation	Keystroke Entry:	**Final Answer:**
5	**Combinations C(5, 2) = 5! / (2! × 3!)** Stack Hint: n choose r	Keystroke Entry:	**Final Answer:**
6	**Combinations C(6, 2) = 6! / (2! × 4!)** Stack Hint: n choose r	Keystroke Entry:	**Final Answer:**
7	**Combinations C(7, 3) = 7! / (3! × 4!)** Stack Hint: n choose r	Keystroke Entry:	**Final Answer:**
8	**Series sum 1+2+3+4+5 via [Σ+]** Stack Hint: Statistical summation	Keystroke Entry:	**Final Answer:**
9	**4! (Factorial)** Stack Hint: x factorial key	Keystroke Entry:	**Final Answer:**
10	**Probability: 1 ÷ C(6, 2)** Stack Hint: Reciprocal combination	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STACKCALC / LEARNING LAB

STUDENT CALCULATOR APPLICATIONS

Unit 12 Time Trial: Digital Logic, Base Conversions

HEX/DEC/OCT/BIN Radix Modes & Bitwise AND, OR, XOR Operations • Stage: Ten-Frame Stage (4x2)

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**DEC 255 to HEX** Stack Hint: Radix mode cycle	Keystroke Entry:	**Final Answer:**
2	**HEX 10 to DEC** Stack Hint: Hex to decimal	Keystroke Entry:	**Final Answer:**
3	**DEC 15 to BIN** Stack Hint: Dec to binary	Keystroke Entry:	**Final Answer:**
4	**BIN 1010 to DEC** Stack Hint: Bin to decimal	Keystroke Entry:	**Final Answer:**
5	**Bitwise AND: 12 AND 10** Stack Hint: 1100 AND 1010 = 1000	Keystroke Entry:	**Final Answer:**
6	**Bitwise OR: 12 OR 10** Stack Hint: 1100 OR 1010 = 1110	Keystroke Entry:	**Final Answer:**
7	**Bitwise XOR: 12 XOR 10** Stack Hint: 1100 XOR 1010 = 0110	Keystroke Entry:	**Final Answer:**
8	**HEX 20 to DEC** Stack Hint: Hex 20 is 32	Keystroke Entry:	**Final Answer:**
9	**DEC 64 to OCT** Stack Hint: Dec 64 is octal 100	Keystroke Entry:	**Final Answer:**
10	**Bitwise AND: 255 AND 15** Stack Hint: Low nybble mask	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

STUDENT CALCULATOR APPLICATIONS

Cumulative Trial C3: Grand RPN Invitational Cham

Master UIL-Style Time Trial across Units 1–12 (Complete Architecture) • Stage: Master Gridfinity Classroom Ecosystem

MAX STACK DEPTH: 4 | **TIME LIMIT: 20 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: ___:___ **Score: ___ / 50**

1	**Compound pipeline: 250 × (1 + .06/12)^24** Stack Hint: TVM compounding	Keystroke Entry:	**Final Answer:**
2	**Pythagorean distance: P1(-3, 5) to P2(5, -10)** Stack Hint: Coordinate lab triple	Keystroke Entry:	**Final Answer:**
3	**Polar vector: 12 cos(45°) + 5 sin(45°)** Stack Hint: Vector projection	Keystroke Entry:	**Final Answer:**
4	**Fractions: 3 1/3 ÷ 1 1/4 + 2/5** Stack Hint: Mixed operations	Keystroke Entry:	**Final Answer:**
5	**Combinations: C(8, 3) = 8! / (3! × 5!)** Stack Hint: Lottery combination	Keystroke Entry:	**Final Answer:**
6	**Exponential: 1500 × e^(0.05 × 10)** Stack Hint: Continuous growth	Keystroke Entry:	**Final Answer:**
7	**Postorder tree: ((35-7)×4)÷(14+2)** Stack Hint: Parenthesis elimination	Keystroke Entry:	**Final Answer:**
8	**Bitmasking: 192 AND 240 (Subnet mask)** Stack Hint: Network bitwise logic	Keystroke Entry:	**Final Answer:**
9	**Double & near-double sum: (6 ENTER +) + (8 ENTE** Stack Hint: Stack auto-lift chain	Keystroke Entry:	**Final Answer:**
10	**Radix conversion: DEC 4095 to HEX** Stack Hint: Maximum 12-bit hex	Keystroke Entry:	**Final Answer:**

UIL Calculator Competition Rules

Scoring: (+5 × Correct) - (2 × Incorrect). Skipped items receive 0. Enforce Max Stack Depth 4 on this trial. Round decimals to 4 significant digits or write exact rational fractions.

TEACHER COMPETITION SOLUTION GUIDE

Official Solutions & Keystrokes — Part 1

Answer keys, required numerical results, and canonical keystroke sequences for Trials 1 to 6 and C1.

Unit 1 Time Trial: Step Counter (Stack 1)

1. Ans: 3 — [1] [ENTER] [1] [+] [1] [+]
2. Ans: 7 — [5] [ENTER] [1] [+] [1] [+]
3. Ans: 4 — [1] [ENTER] [+] [+] [+]
4. Ans: 8 — [1] [ENTER] [+] [+] [+] [+]
5. Ans: 4 — [6] [ENTER] [1] [-] [1] [-]
6. Ans: 5 — [C] [1] [ENTER] [+] [+] [+]
7. Ans: 6 — [2] [ENTER] [2] [+] [2] [+]
8. Ans: 10 — [7] [ENTER] [1] [+] [1] [+]
9. Ans: 9 — [0] [ENTER] [9] [+]
10. Ans: 7 — [10] [ENTER] [1] [-] [1] [-]

Unit 2 Time Trial: Part-Part-Who (Stack 2)

1. Ans: 7 — [4] [ENTER] [3] [+]
2. Ans: 10 — [5] [ENTER] [5] [+]
3. Ans: 3 — [9] [ENTER] [6] [-]
4. Ans: 6 — [10] [ENTER] [4] [-]
5. Ans: 15 — [8] [ENTER] [7] [+]
6. Ans: 6 — [14] [ENTER] [8] [-]
7. Ans: 13 — [6] [ENTER] [7] [+]
8. Ans: 9 — [18] [ENTER] [9] [-]
9. Ans: 20 — [12] [ENTER] [8] [+]
10. Ans: 14 — [25] [ENTER] [11] [-]

Unit 3 Time Trial: Ten-Frames & (Stack 2)

1. Ans: 3 — [10] [ENTER] [7] [-]
2. Ans: 6 — [10] [ENTER] [4] [-]
3. Ans: 13 — [8] [ENTER] [2] [+] [3] [+]
4. Ans: 15 — [9] [ENTER] [1] [+] [5] [+]
5. Ans: 13 — [7] [ENTER] [3] [+] [3] [+]
6. Ans: 9 — [10] [ENTER] [1] [-]
7. Ans: 13 — [20] [ENTER] [7] [-]
8. Ans: 15 — [8] [ENTER] [2] [+] [5] [+]
9. Ans: 16 — [10] [ENTER] [6] [+]
10. Ans: 8 — [10] [ENTER] [2] [-]

Unit 4 Time Trial: Number Racks, (Stack 2)

1. Ans: 12 — [6] [ENTER] [+]
2. Ans: 16 — [8] [ENTER] [+]
3. Ans: 15 — [7] [ENTER] [+] [1] [+]
4. Ans: 13 — [6] [ENTER] [+] [1] [+]
5. Ans: 18 — [9] [ENTER] [+]
6. Ans: 17 — [8] [ENTER] [+] [1] [+]
7. Ans: 12 — [4] [ENTER] [3] [×]
8. Ans: 20 — [5] [ENTER] [4] [×]
9. Ans: 24 — [12] [ENTER] [+]
10. Ans: 19 — [9] [ENTER] [+] [1] [+]

Cumulative Trial C1: Primary Sta (Stack 2)

1. Ans: 7 — [1] [ENTER] [+] [+] [+] [+]
2. Ans: 16 — [9] [ENTER] [7] [+]
3. Ans: 9 — [17] [ENTER] [8] [-]
4. Ans: 7 — [10] [ENTER] [3] [-]
5. Ans: 28 — [14] [ENTER] [+]
6. Ans: 23 — [11] [ENTER] [+] [1] [+]
7. Ans: 17 — [9] [ENTER] [1] [+] [7] [+]
8. Ans: 30 — [5] [ENTER] [6] [×]
9. Ans: 18 — [30] [ENTER] [12] [-]
10. Ans: 21 — [8] [ENTER] [+] [5] [+]

Unit 5 Time Trial: 4-Register St (Stack 4)

1. Ans: 36 — [18] [ENTER] [6] [-] [3] [×]
2. Ans: 5 — [24] [ENTER] [16] [+] [8] [÷]
3. Ans: 5 — [45] [ENTER] [12] [ENTER] [3] [-]
4. Ans: 69 — [15] [ENTER] [4] [×] [18] [ENTER]
5. Ans: 4 — [50] [ENTER] [14] [-] [3] [ENTER]
6. Ans: 28 — [12] [ENTER] [5] [×] [8] [ENTER]
7. Ans: 8 — [20] [ENTER] [8] [-] [4] [×]
8. Ans: 24 — [72] [ENTER] [4] [ENTER] [5] [+]
9. Ans: 32 — [100] [ENTER] [16] [ENTER] [5] [×]
10. Ans: 6 — [8] [ENTER] [4] [+] [9] [ENTER]

Unit 6 Time Trial: Fraction Stud (Stack 4)

1. Ans: 0.75 — [.1.2] [ENTER] [.1.4] [+]
2. Ans: 0.25 — [.3.4] [ENTER] [.1.2] [-]
3. Ans: 3.75 — [2.1.2] [ENTER] [1.1.4] [+]
4. Ans: 1.25 — [.4.5] [1/x]
5. Ans: 0.625 — [.3.8] [ENTER] [.1.4] [+]
6. Ans: 0.5 — [.2.3] [ENTER] [.3.4] [×]
7. Ans: 1.5 — [.3.4] [ENTER] [.1.2] [÷]
8. Ans: 6 — [1.1.2] [ENTER] [4] [×]
9. Ans: 2.25 — [5] [ENTER] [2.3.4] [-]
10. Ans: 6 — [.1.3] [ENTER] [.1.6] [+] [12] [×]

Scoring Adjudication

Accept equivalent decimal fractions (e.g. 0.75 or 3/4). If a student pressed superfluous ENTER keys exceeding the maximum stack depth, deduct 2 points for stack overflow violation.

TEACHER COMPETITION SOLUTION GUIDE

Official Solutions & Keystrokes — Part 2

Answer keys, required numerical results, and canonical keystroke sequences for Trials 7 to 12, C2 & C3.

Unit 7 Time Trial: Expression Tr (Stack 4)

1. Ans: 50 — [8] [ENTER] [2] [+] [9] [ENTER]
2. Ans: 30 — [14] [ENTER] [6] [-] [3] [×]
3. Ans: 20 — [30] [ENTER] [4] [ENTER] [5] [×]
4. Ans: 12 — [5] [ENTER] [3] [+] [8] [ENTER]
5. Ans: 40 — [18] [ENTER] [3] [÷] [4] [+]
6. Ans: 5 — [10] [ENTER] [5] [ENTER] [2] [×]
7. Ans: 70 — [25] [ENTER] [5] [ENTER] [3] [×]
8. Ans: 8 — [7] [ENTER] [3] [+] [4] [×]
9. Ans: 36 — [48] [ENTER] [2] [ENTER] [6] [+]
10. Ans: 30 — [16] [ENTER] [4] [ENTER] [2] [+]

Unit 8 Time Trial: Coordinate La (Stack 4)

1. Ans: 5 — [3] [x²] [4] [x²] [+] [√□]
2. Ans: 10 — [6] [x²] [8] [x²] [+] [√□]
3. Ans: 13 — [5] [x²] [12] [x²] [+] [√□]
4. Ans: 5 — [4] [ENTER] [1] [-] [x²] [6]
5. Ans: 10 — [8] [ENTER] [2] [-] [x²] [11]
6. Ans: 3 — [12] [ENTER] [4] [÷]
7. Ans: 3 — [17] [ENTER] [5] [-] [6] [ENTER]
8. Ans: 15 — [9] [x²] [12] [x²] [+] [√□]
9. Ans: 17 — [8] [x²] [15] [x²] [+] [√□]
10. Ans: 25 — [7] [x²] [24] [x²] [+] [√□]

Cumulative Trial C2: Intermediat (Stack 4)

1. Ans: 15 — [32] [ENTER] [14] [-] [5] [×]
2. Ans: 0.875 — [.1.2] [ENTER] [.1.4] [×] [.3.4] [
3. Ans: 10 — [9] [ENTER] [3] [-] [x²] [12]
4. Ans: 1.75 — [3.1.2] [ENTER] [1.3.4] [-]
5. Ans: 10 — [15] [ENTER] [5] [+] [8] [ENTER]
6. Ans: 27 — [18] [ENTER] [+] [9] [-]
7. Ans: 3 — [10] [ENTER] [2] [+/-] [-] [3]
8. Ans: 32 — [80] [ENTER] [15] [ENTER] [4] [×]
9. Ans: 2.4 — [.1.4] [ENTER] [.1.6] [+] [1/x]
10. Ans: 20 — [12] [x²] [16] [x²] [+] [√□]

Unit 9 Time Trial: Trigonometry, (Stack 4)

1. Ans: 0.5 — [30] [SIN]
2. Ans: 0.5 — [60] [COS]
3. Ans: 1 — [45] [TAN]
4. Ans: 5 — [10] [ENTER] [30] [SIN] [×]
5. Ans: 6 — [12] [ENTER] [60] [COS] [×]
6. Ans: 10 — [10] [ENTER] [0] [yellow] [□y,x]
7. Ans: 5 — [3] [ENTER] [4] [yellow] [□□,r]
8. Ans: 13 — [5] [ENTER] [12] [yellow] [□□,r]
9. Ans: 10 — [8] [ENTER] [6] [yellow] [□□,r]
10. Ans: 30 — [.5] [yellow] [ASIN]

Unit 10 Time Trial: Exponential (Stack 4)

1. Ans: 32 — [2] [ENTER] [5] [y□]
2. Ans: 81 — [3] [ENTER] [4] [y□]
3. Ans: 1 — [1] [e□] [LN]
4. Ans: 1 — [0] [e□]
5. Ans: 3 — [1000] [yellow] [LOG]
6. Ans: 1102.5 — [1000] [ENTER] [1.05] [ENTER] [2]
7. Ans: 9.9021 — [2] [LN] [.07] [÷]
8. Ans: 12.5 — [100] [ENTER] [.5] [ENTER] [3] [y□
9. Ans: 610.7014 — [.04] [ENTER] [5] [×] [e□] [500]
10. Ans: 1024 — [2] [ENTER] [10] [y□]

Unit 11 Time Trial: Combinatoric (Stack 4)

1. Ans: 120 — [5] [yellow] [x!]
2. Ans: 720 — [6] [yellow] [x!]
3. Ans: 20 — [5] [yellow] [x!] [3] [yellow] [x!
4. Ans: 120 — [6] [yellow] [x!] [3] [yellow] [x!
5. Ans: 10 — [5] [yellow] [x!] [2] [yellow] [x!
6. Ans: 15 — [6] [yellow] [x!] [2] [yellow] [x!
7. Ans: 35 — [7] [yellow] [x!] [3] [yellow] [x!
8. Ans: 15 — [1] [□+] [2] [□+] [3] [□+]
9. Ans: 24 — [4] [yellow] [x!]
10. Ans: 0.0667 — [6] [yellow] [x!] [2] [yellow] [x!

Unit 12 Time Trial: Digital Logi (Stack 4)

1. Ans: FF — [255] [yellow] [BASE] [HEX]
2. Ans: 16 — [HEX] [10] [yellow] [BASE] [DEC]
3. Ans: 1111 — [15] [yellow] [BASE] [BIN]
4. Ans: 10 — [BIN] [1010] [yellow] [BASE] [DEC]
5. Ans: 8 — [12] [ENTER] [10] [AND]
6. Ans: 14 — [12] [ENTER] [10] [OR]
7. Ans: 6 — [12] [ENTER] [10] [XOR]
8. Ans: 32 — [HEX] [20] [yellow] [BASE] [DEC]
9. Ans: 100 — [64] [yellow] [BASE] [OCT]
10. Ans: 15 — [255] [ENTER] [15] [AND]

Cumulative Trial C3: Grand RPN I (Stack 4)

1. Ans: 281.79 — [.06] [ENTER] [12] [÷] [1] [+]
2. Ans: 17 — [5] [ENTER] [3] [+/-] [-] [x²]
3. Ans: 12.0208 — [12] [ENTER] [45] [COS] [×] [5]
4. Ans: 3.0667 — [3.1.3] [ENTER] [1.1.4] [÷] [.2.5]
5. Ans: 56 — [8] [yellow] [x!] [3] [yellow] [x!
6. Ans: 2473.08 — [.05] [ENTER] [10] [×] [e□] [1500]
7. Ans: 7 — [35] [ENTER] [7] [-] [4] [×]
8. Ans: 192 — [192] [ENTER] [240] [AND]
9. Ans: 29 — [6] [ENTER] [+] [8] [ENTER] [+]
10. Ans: FFF — [4095] [yellow] [BASE] [HEX]

Scoring Adjudication

Accept equivalent scientific notation for large magnitudes (e.g. 1024 or 1.024e3). All answers match canonical RPNCore engine execution.